IMAGE BROWSING WITH PCA-ASSISTED USER-INTERACTION

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ABSTRACT

User interfaces for sophisticated search engines with relevance feedback must offer users a quick and easy access to the images to be visualized. We present a user interface, where image features are described in terms of very high-dimensional MPEG-7 descriptors and are reduced to only three dimensions for visual presentation afterwards. The dimension reduction is realized by an appropriate weighting of the high-dimensional descriptor components corresponding to a modification of the covariance-matrix used for Principal Component Analysis (PCA). In addition, this modification allows to overcome a problem arising from equally sized Eigenvalues and provides varying Eigenspaces nearly continousely.
The technique introduced is a general approach, which can be combined with other relevance feedback methods. The user can adapt the visualization procedure according to his personal preferences in a continuous and intuitive manner in real time.

Keywords: relevance feedback, image browsing, user-interface, shape-based MPEG-7, realtime

1 INTRODUCTION

The importance of user interfaces for various kinds of search engines is growing rapidly. Applications for this search engines are ranging from image databases over Video-On-Demand systems to the whole field of E-Commerce systems.
To be successful, user interfaces must present objects in an appealing manner, allow easy and intuitive navigation through the set of objects and highlight their characteristics. This can be done very successfully by taking into account the very high recognition capability of the human visual system. Visual similarity and visual order are major factors in searching. If images are being arranged with respect to their similarity instead of using the currently widespread list presentation, the exhaustion during the search will be avoided. In addition, the user can mark intuitively visual areas matching his interest and can refine his search in these areas.
The MPEG-7 descriptors for still image- and video description are well adapted to visual feature extraction and are also suited for fast indexing. Each descriptor covers a part of a high dimensional feature space, which may achieve up to 200 or even more dimensions.
But for visualization the feature space has to be reduced to 2 or 3 dimensions. We use Principal Component Analysis (PCA) here for dimension reduction. The computation of the covariance matrix is a time consuming procedure, but it has to be done only once after the database has been created. So its computation does not affect the search engine at run time.
The PCA yields an uncorrelated feature vector representation with minimum difference between the original feature vector and the feature vector, which is reconstructed from the dimensionally reduced one [2]. This is achieved keeping only those components of the transformed feature vector having the highest variance. The PCA provides the statistical optimum in the sense of Least Squares [3].
However, components being salient in a pure statistical sense might not be relevant to the user. Rather the user may be interested in some particular aspects with minor statistical importance. Therefore, a user-specific weighting of feature components is needed to take into account personal preferences.
In contrast to different interactive systems our user interface enables fast visualization, easy use and intuitive presentation. MARS [14] for example, asks the user to rate the search results. PicSOM [10] uses Learning Vector Quantization for the dimension reduction, and provides, apart from a plain relevance feedback technique, no explicit possibilities for an intervention by the user. In [10] a system is introduced which applies learning algorithms for data, task and context recognition. Although appealing, these approaches are somewhat circumstantial and not suited for easy and quick application.
With interactive systems, the user permanently reacts to system output and permanently changes user-specific parameters. And with the linear compression technique proposed here, user interests can be integrated by weighting each feature vector component by a factor expressing the user's preference followed by a new computation of the entire PCA.

2 DIMENSION REDUCTION

As described above, the first step is dimension reduction. Non-linear techniques like Neural Networks (especially Multilayer Perceptrons and Self Organizing Maps [3], [11] [1]) provide a better adaption to often heavily folded clusters in the feature space compared to the PCA. However, the integration and adjustment of user-specific weights is hard to realize. Therefore, PCA is used here, as already introduced in [9].

The PCA rotates and shifts the coordinate system in such a way that features are made uncorrelated. After this, feature components with small variance are neglected. This corresponds to a projection into a low dimensional subspace capturing most of the variation within the data set [8].
The PCA (for details see [2], [3], [7]) takes a set of real n-dimensional feature vectors as input. Let $x_1,\dots, x_N$ denote the feature vectors, which result from the original ones after substraction of their mean. After this, the covariance matrix C is the expectation of the matrix $X_{ij} := x_i \cdot x_j$, i.e.

$$C = \mathrm{E}(X) \tag{1}$$

Lets $v_1,\dots, v_n$ be the normalized eigenvectors of C, arranged in decreasing order of their eigenvalues $\lambda(1),\dots, \lambda(n)$, and define the projection matrix V to be an $n \times k$ matrix having the eigenvectors $v_1,\dots, v_k$ as columns. The vectors $v_1,\dots, v_n$ are an orthonormal basis of the n-dimensional Euclidean feature space. For each original feature vector x a k-dimensional feature vector y is computed by application of the transposed matrix of V to x, i.e.

$$y = V^T x \tag{2}$$

$$\text{with } k \ll n,\ k \in \{2,3\} \text{ here.} \tag{3}$$

The Eigenvalues $\lambda(1),\dots, \lambda(k)$ are the variances of the new feature vectors y. Note, that y is a vector of a reduced number of dimensions corresponding to the largest variances.

2.1 Integration of user-intention

There are two possibilities to integrate user preference weights into the PCA. The first one virtually changes the variance of the data by rescaling the axes. After this is done the covariance matrix C must be computed again. Alternatively, only the columns and rows concerned are adjusted. Suppose, the m-th component of all zero mean feature vectors $x_1,\dots, x_N$ shall be weighted by a factor α. This corresponds to an increment of the variance of dimension m. Then yields

$$x'_n(m) := \alpha \cdot x_n(m),\ \forall n \in \{1,\dots,N\}. \tag{4}$$

However, this corresponds straightforward to a modification C' of the covariance matrix C as:

$$C'(i,j) = \begin{cases} C(i,j), & i,j \neq m \\ \alpha \cdot C(i,j), & i = m \vee j = m \\ \alpha^2 \cdot C(i,j), & i = j = m \end{cases} \tag{5}$$

Due to eq. (5) the recomputation of the covariance matrix can be avoided. Instead only some columns and rows are multiplied by weights. However, this still requires a new computation of eigenvectors and eigenvalues.
Note, that a normalization in (5) with respect to a total sum of variances is not performed. This complies with the user intention, who wants to spread out a certain detail.
It is obvious from the formulas above, that by the introduced approach user preferences control the principal components chosen within the PCA. This enables the user to visualize those subspaces which appear relevant to him.

3 EQUALY SIZED EIGENVALUES

At least k Eigenvalues have to be derived from the covariance matrix C. Two or even more of them achieving comparable magnitude cause a problem. Then their order may change, and Eigenvalues, not considered before now belong to the highest three ones with the highest magnitude. In this case, their dedicated Eigenvector also belongs to the visualized subspace. Usually, the newer Eigenvector is orthogonal to the older one, yielding to a complete distortion of the visualization. This problem is non trivial to overcome, but with the dimension scaling (5) introduced, there is a non-analytic way.
Note, that the prime order of Eigenvalues depends on the algorithm used for the extraction of zero crossings of the characteristic polynomial. There is no direct connection to the origin. Commonly used here are the Householder or the Jacobi transform ([13], [4]).
Consider a simple example. Assume a two-dimensional data set, containing uncorrelated normal distributed data with zero mean and variance set to 1. Now, the data will be scaled acc. to eq. (5), as:

$$x_1 := x_1^N \cdot \alpha, \text{ and } x_2 := x_2^N \cdot \frac{1}{\alpha}, \tag{6}$$

where the N describes normal distributed data. Let α vary arbitrarily between

$$\alpha \in \{0.5; 1.5\}, \tag{7}$$

as depicted in the Figure 2.

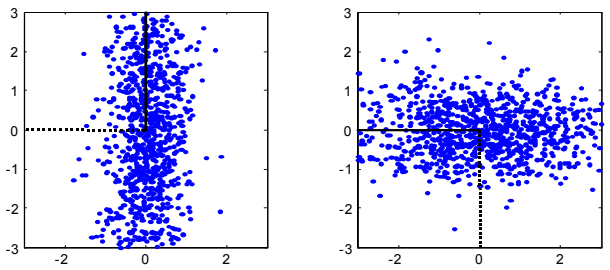


Fig. 2: Scaling of a 2-dim. distribution acc. to eq. (6), (7) and both Eigenvectors; left: $\alpha = 0.5$; right: $\alpha = 1.5$

Note, the samples themselves have not been changed. However the first principal vector has been turned to the

orthogonal direction. This effect becomes still more serious if a dimension reduction is applied. Then, the entire projection axis will rotate.
Note further, the factor for axis scaling is proportional to the scaling of the standard deviation in this dimension and is equally to the expression of preference according to eq. (5).
Applying a continuously shifting of a, we receive the continuous behaviour of the Orinetation of Eigenvectors, depicted in Figure 3.

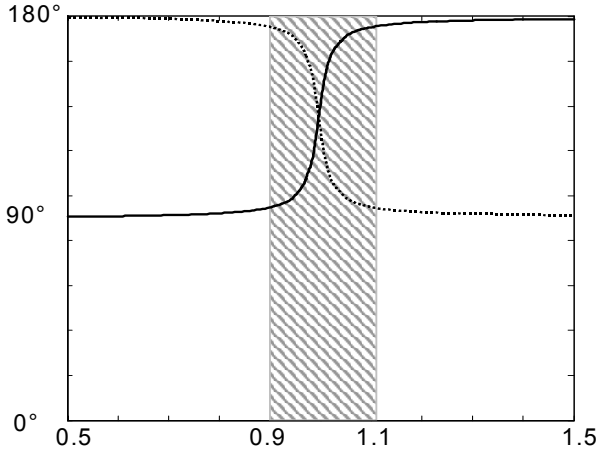


Fig. 3: Orientation of the principal axes with respect to the abscissa over α ; gray area: "sensible" domain

If the steps changing α do not become too large, Fig. 3 shows a behaviour smooth enough for a slow transition between both stable states. Abscissa values in the range {0.9, … 1.111} causes the angle transition, and we call this range "sensitive". Skipping over the sensitive area, occurring by a step too large, yields the disturbing effect of the subspace. The problem growths rapidly preserving only the distribution instead of the samples themselves.
In our example, the dedicated Eigenvalues act proportional to the variance of the input vectors, since these are uncorrelated. At the points {0.9, 1.111} they achieve a magnitude of 0.81 and 1.23, resp. Both Eigenvalues of this special example are depicted in Fig. 4.

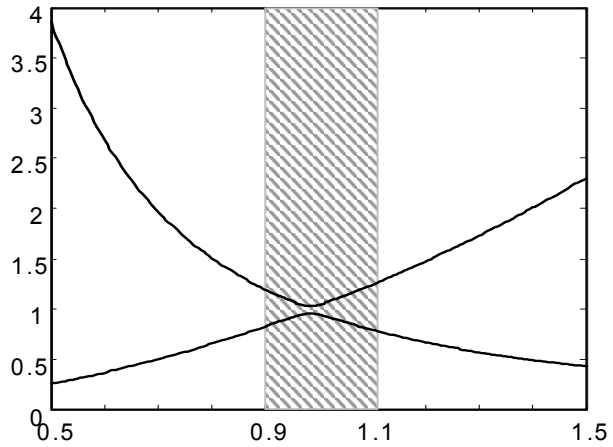


Fig. 4: Eigenvalues of example Fig. 2

Note, that in our example both Eigenvalues never achieve the value of 1 at the same time. This due to the algorithm used for the Eigenvalue computation [4], and is the assumption for yielding the quasi-continuous depiction of Fig. 3. Other algorithms yielding exact same sized Eigenvalues, would lead to a multiple of Eigenvectors at this point.
The normalized difference

$$\Delta\lambda_{ij} := \frac{\lambda(i) - \lambda(j)}{\lambda(i) + \lambda(j)}, \tag{8}$$

of both Eigenvalues λ_i and λ_I leads to the range

$$\left|\Delta\lambda_{ij}\right| \le 0.205 \qquad \textbf{sensible domain} \tag{9}$$

being a "sensible" domain. The value is derived form the assumptions made above.

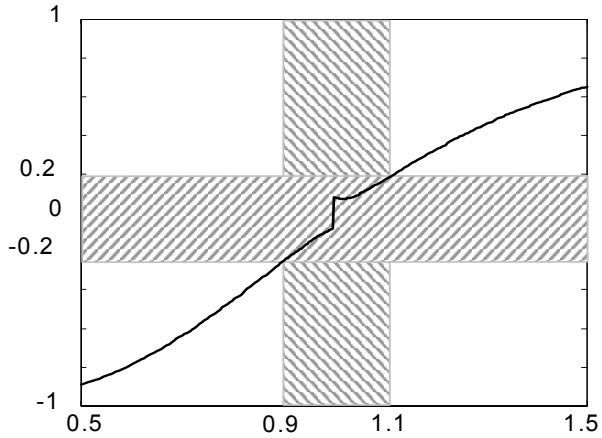


Fig. 5: Normalized difference of Eigenvalues (8) over α

The discontinuity near the origin in Fig. 5 is due to the two-fold zero crossings close to 1.
Finally, we introduce the depiction of Eigenvector orientation over the normalized Eigenvalues, as shown in Fig. 6.

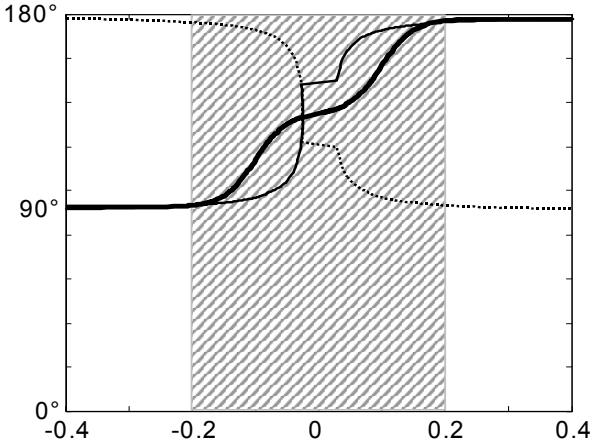


Fig. 6: Eigenvector orientation (Fig. 3) over the normalized difference of Eigenvalues (8), gray: sensitive domain; thick line: rotation replacement function

This depiction has the advantage to be valid in general for each 2D subspace spanned by two Eigenvectors. The discontinuity in the middle of Fig. 6 is due to the two-fold zero crossing as described at Fig. 5.
Avoiding all problems resulting from steps too big, multiple zero crossings, or from changing samples, we suggest using a rotation replacement function for. This function should perform the rotation between both borders of the sensible area, but has zero slope near the discontinuity shown in Fig. 6. We suggest

$$f(\Delta\lambda_{ij}) = \left[\text{tansig}\left((\Delta\lambda_{ij} + 0.1)\cdot 20\right) + \text{tansig}\left((\Delta\lambda_{ij} - 0.1)\cdot 20\right)\right]\cdot 22.5° + 135° \tag{10}$$

Hereby, tansig(x) means the tangens-Sigmoid function commonly used in Neural Networks. With

$$\text{tansig}(x) := \frac{2}{1+\exp(-2x)} - 1 \tag{11}$$

this yields

$$f(\Delta\lambda_{ij}) = \left[\frac{1}{1+\exp(-40\Delta\lambda_{ij} - 4)} + \frac{1}{1+\exp(-40\Delta\lambda_{ij} + 4)} - 1\right]\cdot 45° + 135°$$

as rotation replacement function, shown in Fig. 6. This is a monotonic function which avoids angle shifts in the middle of the sensible range.
For practical use, all

$$P(k) = \binom{k}{2}, \quad \text{with } k = \{5,\ldots,8\} \tag{12}$$

pairs of the highest k Eigenvalues have to be tested to span a *sensible* 2D subspace. Doing this, the "unsensible" Eigenvectors have to be derived, and the rotation replacement function is to be applied.
In this way, certain 2D subspaces may be rotated artificially and not to perform a statistical optimum. Still, this preserves smooth transitions during the user-specific visualization.

4 EXAMPLES OF USER-SPECIFIC SEARCH

The efficiency of the user-interactive system is demonstrated for a database storing about 3500 different images of shapes. It has been provided by the MPEG-7 community and is commonly used for the evaluation of descriptors.
As an example, we have extracted two shape-based MPEG-7 descriptors: *BoundingBox* and *ContourBasedShape* ([7], [15]). Because these descriptors primarily have



Fig. 7a,7b,7c: Continuous transition of feature weightings:
a) (top): *BoundingBox*
b) 50% *BoundingBox* and 50% *ContourBasedShape*
c) (bottom): *ContourBasedShape*

been developed for matching by pairs, they have to be transformed several times until they span an Euclidian vector space, finally being invariant with respect to translation, rotation, scaling, and mirroring. This yields a 60-dimensional space which has to be projected into a three dimensional subspace suitable for visualization.

We prefer 3-dimensional visualization due to the more improved cognitive capabilities of the human visual system in space over 2-dimensional plane. The 3D effect is achieved by a scene rotation, which causes a strong spatial illusion.

By means of a few sliders the user is now able to control search items and subspace axis in the high-dimensional feature space corresponding to "orientation" and "contour". These sliders adjust user preference weights and correspond to the rescaling of the original feature space spanned by the MPEG-7 descriptors. In principal, any feature extraction algorithm can be integrated into this visualization tool provided the features can be considered elements of some vector space.

For an example, in Fig. 7a … 7c, we demonstrate the continuous transition from one subspace visualization into another. At the first glance the user might concentrate him self to the orientation of objects and to the ratio of the object's width to height. He will pay thus most attention to visual aspects being captured by the *BoundingBox* descriptor, as shown in Fig. 7a. Than the user might prefer an alignment, emphasizing variations of the contour, as shown in Fig. 7c. He would pay more attention to the capacity of the *ContourShape* descriptor. Fig. 7b reflects a gradual shift of the user's preferences starting at shape proportion and shape orientation and arriving at a point of view guided by contour complexity in the end. This transition is performed in real time.

5 CONCLUSION

We have shown that PCA can be applied for the visulization of high-dimensional MPEG-7 descriptors. It yields an elegant method for effective user-interaction for search engines. Though the PCA is a linear technique, its performance is powerful and sufficient for the compression of high dimensional feature spaces to only 3 dimensions.

Usually, the introduction of a user-specific weighting will require a time consuming recomputation of the PCA's covariance matrix. However, we found quite an easy way to circumvent this problem.

Strongly connected to the continuously varying visualization is the problem of having two Eigenvalues with the same magnitude. We overcome this by introducing a suitable rotation replacement function.

Hence, our user interface allows a continuous variation of the visualized subspace in accordance with the search preferences of the user.

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